

3.11.1 Rotational dynamics (A-level only)

3.11.1.1 Concept of moment of inertia (A-level only)

Content

$I = mr^2$ for a point mass. $I = \sum mr^2$ for an extended object.

Qualitative knowledge of the factors that affect the moment of inertia of a rotating object.

Expressions for moment of inertia will be given where necessary.

$$E_k = \frac{1}{2} I \omega^2$$

Factors affecting the energy storage capacity of a flywheel.

Use of flywheels in machines.

Use of flywheels for smoothing torque and speed, and for storing energy in vehicles, and in machines used for production processes.

3.11.1.3 Rotational motion (A-level only)

Content

Angular displacement, angular speed, angular velocity, angular acceleration, $\omega = \frac{\Delta \theta}{\Delta t}$, $\alpha = \frac{\Delta \omega}{\Delta t}$

Representation by graphical methods of uniform and non-uniform angular acceleration.

Equations for uniform angular acceleration;

$$\omega_2 = \omega_1 + \alpha t, \theta = \left(\frac{\omega_1 + \omega_2}{2} \right) t$$

$$\theta = \omega_1 t + \frac{\alpha t^2}{2}, \omega_2^2 = \omega_1^2 + 2\alpha\theta$$

Students should be aware of the analogy between rotational and translational dynamics.

3.11.1.4 Torque and angular acceleration (A-level only)

Content

$$T = Fr$$

$$T = I\alpha$$

3.11.1.5 Angular momentum (A-level only)

Content

$$\text{angular momentum} = I\omega$$

Conservation of angular momentum.

Angular impulse = change in angular momentum; $T \Delta t = \Delta(I\omega)$ where T is constant.

Applications may include examples from sport.

$$W = T\theta; P = T\omega$$

Awareness that frictional torque has to be taken into account in rotating machinery.